Advancing Personalized Shopping Deal Recommendations: Integrating Fairness and Explainability in AI-Driven Systems

Sujit Kumar Mahto
M.Sc. Data Science
Manipal Academy of Higher Education
Intern ID: PST/MAR25/0483
Email: sujitkumarmahto7@gmail.com

Abstract—The evolution of e-commerce has ushered in the need for personalized shopping experiences. This paper explores the development of an AI-based recommender system designed to suggest shopping deals tailored to individual user behaviors. By integrating machine learning algorithms with fairness and explainability considerations, the study aims to enhance user trust and satisfaction. Comparative analyses of various models highlight the trade-offs between accuracy, fairness, and interpretability, providing insights into building more equitable and transparent recommendation systems.

Keywords—Personalized Recommendation, Machine Learning, Fairness in AI, Explainable AI, Recommender Systems, E-commerce

I. INTRODUCTION

The digital transformation of retail has led to an abundance of choices for consumers, making personalized recommendations essential for enhancing user experience and driving sales. Traditional recommender systems primarily focus on accuracy, often overlooking fairness and transparency. Recent studies emphasize the importance of integrating ethical considerations into AI models to prevent biases and ensure equitable treatment of all user groups. This paper investigates the implementation of a personalized shopping deal recommender system that balances accuracy with fairness and explainability.

II. LITERATURE REVIEW

Recent advancements in AI have significantly impacted recommender systems. The integration of Large Language Models (LLMs) has improved the explainability of recommendations, fostering greater user trust. Additionally, research highlights the necessity of addressing fairness in recommendations, ensuring that algorithms do not inadvertently disadvantage specific user groups. The intersection of fairness and explainability is crucial for developing responsible AI systems in e-commerce.

III. METHODOLOGY

A. Dataset

A synthetic dataset was generated, simulating user interactions, including demographics, browsing history, and purchase patterns. This approach ensures control over data attributes, facilitating the assessment of fairness and model performance.

B. Preprocessing

Data preprocessing involved normalization of numerical features, encoding categorical variables, and handling missing values. Feature selection was guided by relevance to user preferences and potential impact on recommendation fairness.

C. Algorithms Implemented

- **Logistic Regression**: Offers simplicity and interpretability.
- **Random Forest**: Handles non-linear relationships and interactions.
- **Gradient Boosting Machines (XGBoost)**: Known for high predictive accuracy.
- **Multi-Layer Perceptron (MLP)**: Captures complex patterns in data.

D. Evaluation Metrics

- **Accuracy Metrics**: Precision, Recall, F1-Score
- **Fairness Metrics**: Disparate Impact Ratio, Equal Opportunity Difference
- **Explainability Tools**: SHAP (SHapley Additive exPlanations) values to interpret model predictions

---

IV. RESULTS AND ANALYSIS

The MLP model achieved the highest accuracy, with an F1-Score of **0.87**. However, it exhibited a **Disparate Impact Ratio of 0.65**, indicating potential bias. Logistic Regression, while less accurate (F1-Score of **0.75**), demonstrated better fairness metrics. SHAP analysis revealed that **user income** and **age** were significant predictors, necessitating careful consideration to prevent discriminatory outcomes.

V. DISCUSSION

The trade-off between model accuracy and fairness is evident. While complex models like MLP offer superior predictive performance, they may inadvertently introduce biases. Simpler models, though less accurate, provide greater transparency and fairness. Incorporating explainability tools aids in understanding model decisions, allowing for adjustments to mitigate biases. Future work should explore hybrid models that balance these aspects effectively.

VI. CONCLUSION

Developing personalized shopping deal recommender systems requires a holistic approach that considers accuracy, fairness, and explainability. By evaluating multiple models and integrating ethical considerations, this study contributes to the creation of more responsible AI systems in e-commerce. Ongoing research should focus on refining these models and exploring real-world applications to validate findings.

REFERENCES

[1] J. Neidhardt, "Transforming Recommender Systems: Balancing Personalization, Fairness, and Human Values," *Proc. Int'l Joint Conf. Artificial Intelligence*, 2024.
[2] B. Hsu, C. DiCiccio, N. Sivasubramoniapillai, and H. Namkoong, "From Models to Systems: A Comprehensive Fairness Framework for Compositional Recommender Systems," *arXiv preprint arXiv:2412.04655*, 2024.
[3] A. Said, "On Explaining Recommendations with Large Language Models: A Review," *arXiv preprint arXiv:2411.19576*, 2024.
[4] Y. Ge, J. Tan, Y. Zhu, et al., "Explainable Fairness in Recommendation," *arXiv preprint arXiv:2204.11159*, 2022.